



DEALER BEST PRACTICE SERIES

Dyno Room Fluid Lines

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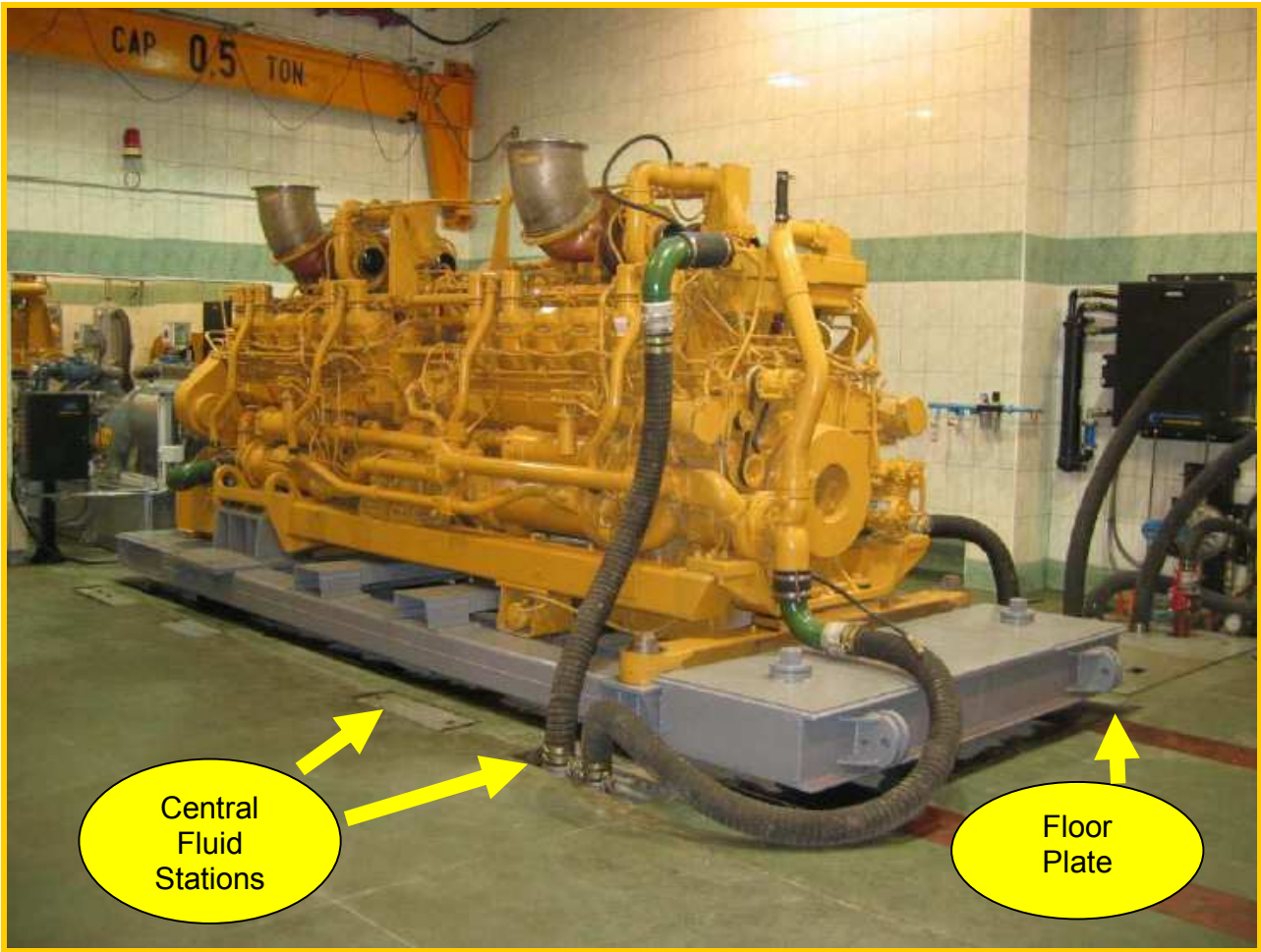
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1.0 Introduction

Many engine dynamometer test cells are cluttered with an array of fluid hoses routed to/from the cell wall, the cooling column(s), and across the floor to the engine. Large hoses lying across the floor interfere with mechanics and other test cell personnel activities and hamper access to the engine during and after test. An effective solution is to install fluid supply lines under the dyno room floor to terminate at a central fluid station next to tested engine.

2.0 Best Practice Description



An engine dynamometer test cell with hoses routed under the floor

3.0 Implementation Steps

Note: Reference Caterpillar publication, [\(archived\) SEBD0642-02, Engine Dynamometer Facilities](#), and/or the Dyno Supplier, for specific test cell utility requirements before changing facility arrangements.

This practice is relatively simple and inexpensive to implement.

1. The first step would be to identify the ideal spot along side the typical test engines to locate a central fluid station. Some dealers have installed several central fluid stations to match the size of different engine models.
2. Identify the ideal spot to mount the cooling column against the test cell wall.
3. Next, design the best trough routing for the fluid supply lines from the source to the central fluid station. Measure the distance and fabricate the fluid lines to be installed in the floor. Insure routing plan will not cause air pockets in the lines.
4. Construct a shallow channel in the floor deep enough to contain the fluid lines and still be easy to access for maintenance or repair. Install a waste drain in the trough if possible. Design the channel so that a removable steel plate or grate can be installed to cover the channel and fluid lines. Grate or plate should sit flush with the shop floor so that it does not protrude or recess from the floor level.
5. Construct the cooling column support. If it must be mounted on the floor, use the stand for storage that will not collect floor dirt and need to be cleaned underneath. Install the hoses/lines that will lead to the cool water source/discharge and to the engine connections.
6. Design and fabricate a central fluid station. The fluid station is usually a simple metal box with permanent fluid supply lines entering the box and attached to quick connect fittings. It is also advisable to install a shut off valve for each fluid supply line.
7. Fabricate fluid hoses with quick connect couplings to provide fluid from the central fluid station(s) to the engine. Length/size needed, will vary by engine model.

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4.0 Benefits

1. **Security.** With fluid hoses under the floor, there is less chance of worker injury from a trip and fall incident. Paths around the dyno/engine are unobstructed for quick personnel movement during standard duties and for emergency response/evacuation.
2. **Efficiency.** With the fluid supply line under the floor, technicians have easy access to the engine for installation, test and adjustment, and removal. The process is more efficient when there are no large hoses or cooling columns in the way to hamper movement. The cooling tower to cooling column hoses would be permanently attached, and only the engine to cooling column hoses need adapter connections. This would reduce cooling system connection adapters.
3. **Clean up.** With central fluid stations and lines routed under the floor, the chance of spillage is lower and clean up time significantly reduced. Connection spillage is often contained/drained through the trough if drains are installed. Wall mounted cooling columns eliminate some connections and do not have to be moved to clean under them.

5.0 Resources Required

The investments required would involve constructing an in-floor trough to route fluid supply/return lines from the cooling column and fuel supply to a central fluid station adjacent to the tested engine. The trough in the floor would require a metal grate or plate cover. The fluid station is a simple panel of quick connect fittings for the fuel and coolant supply lines. Cooling columns would be wall-mounted in the engine front end of the test cell. Total investment is usually under \$10,000 US.

6.0 Supporting Attachments & References

See also:

Caterpillar publication [\(archived\) SEBD0642-02 Engine Dynamometer Facilities](#)

7.0 Related Best Practices

Remote Control Cameras in Dyno Room Improve Safety	0408-5.1-1118
Plug and Play System for Engine Readiness in the Dynamometer	1605-4.03-1435
Connection Bypass on Main Cooling for 3500 Engine Test	1605-4.03-1434
Kits for Dynamometer Test	1408-4.01-1353

8.0 Acknowledgements

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